



Analysis of meteorological parameters for dense gas dispersion using mesoscale models



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ABSTRACT

The current study focuses on characterizing the atmospheric details required for dense gas dispersion analysis resulting from release of cryogenic liquids like LNG. The study investigates the effectiveness of coupling the prognostic MM5 mesoscale model with the CALMET diagnostic model for producing meteorological conditions that is characteristic of dry and arid regions like Qatar, with non-neutral boundary conditions. MM5-CALMET wind fields and temperature data are compared with the meteorological field observations from the Doha International Airport (DIA) on a monthly basis, daily basis and hourly basis to study the effect of different averaging periods. The monthly averages replicate the annual patterns of meteorological parameters very well. However difference in observation and model are observed for wind speed and wind direction variable. The daily averages obtained from the model are in good agreement with the observation for wind speed and temperature. For hourly averaging, the model is found capable of mimicking the temperature of a given location, but not wind speed and direction. The prediction of wind direction parameter using MM5-CALMET is moderate for any averaging period. The sub-optimal performance of wind direction variable is attributed to grid resolution of vertical and horizontal layers of MM5-CALMET model. Additionally a case study is performed to illustrate the effect of variation of meteorological parameters on the lower flammability limits (LFL) resulting from flammable dense gas release of LNG. The case study demonstrates the issues that arise in a risk analysis study when “wrong” meteorological data could be used. The overall study indicates that utilizing the coarsest prognostic meteorological model output in a diagnostic model provides an effective option for generating meteorological inputs for dispersion studies.

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1. Introduction

The basic approach for constructing the atmospheric conditions for dispersion typically utilizes either a prognostic model or a diagnostic model. Diagnostic models take into account the available observations and topography of terrain (Scire et al., 2000). Since the diagnostic models interpolate observational data, they are critically dependent on the quality and the density of these observations. Owing to this reason, diagnostic models cannot be used for forecasting purposes. Prognostic

meteorological models solve primitive equations that govern the state of the atmosphere (Pielke and Uliasz, 1998). These models solve the equations of motion in time and space and can be used for forecasting. Mesoscale models are prognostic models that predict the meteorological behavior at horizontal scales ranging from few kilometers to few hundred meters. Examples of selected mesoscale models include MM5 (Dudhia et al., 2005), HOTMAC (Yamada and Henmi, 1994) and RAMS (R. A. Pielke et al., 1992). Among the mesoscale models the MM5 model is well-established and has been widely applied for different kinds of meteorological studies (Ghannam and El-Fadel, 2013). The problem with the fully prognostic model is the uncertainties in inputs and the simplifying assumptions in the formulations that can cause predictions which do not match actual observations.

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Currently there is an increase in the use of combined prognostic and diagnostic models, which complement their individual advantages (Chandrasekar et al., 2003; Ghannam and El-Fadel, 2013). Robe and Scire (Robe and Scire, 1998) initially proposed to combine coarse prognostic model with diagnostic model of fine resolution. They evaluated the model in terms of surface wind flow for brief meteorological period. Barna et al. (2000) simulated the meteorological fields for ozone modeling. They had used MM5 and CALMET combination with grid spacing of 5 km. Such studies concluded that the success of coupled prognostic/diagnostic modeling approach lied in the validation with actual field observations.

The present study investigates the effectiveness of coupling the prognostic MM5 mesoscale model with the CALMET diagnostic model for producing meteorological conditions that are characteristic of Qatar's atmospheric environment. Qatar has a dry and arid climate with non-neutral meteorological conditions. Typical risk analysis studies emphasize the dispersion outcomes in neutral atmospheric conditions. However, it is to be observed that majority of the industrial sites exists in non-neutral atmospheric conditions. For risk analysis studies, the meteorological input is generally provided either from the Doha International Airport station or most often from MM5 model data single-point extracts. The meteorological stations of industrial cities do not always contain appropriate source of data either because of lack of proper Quality Assurance/Quality Control (QA/QC), low availability or the additional requirement of post-processing in order to be used in selected modeling systems, like the EPA recommended dispersion modeling system AERMOD (EPA, 2004). AERMOD requires surface hourly winds and temperature, hourly cloud cover and morning soundings of wind, temperature and dew point from the nearest available observation site. The data are processed using a software routine called AERMET which calculates the hourly turbulence parameters, friction velocity, mixing height, Monin Obukhov length and other boundary layer parameters. It is to be noted that for AERMOD, the necessary surface and upper air meteorological data are usually retrieved from the closer single point of the "coarse" MM5 grid (usually 12 km or 4 km). It operates under an assumption of uniform meteorological condition. This is one of the primary reasons for choosing the diagnostic model CALMET to estimate meteorological predictions with highest resolution for near and far field applications. The outputs of CALMET are passed onto CALPUFF which is a lagrangian model. The lagrangian model in CALPUFF routine has the capability to simulate dispersion in non-uniform, spatially varying meteorological conditions.

The current study specifically focuses on characterizing the atmospheric details required for dense gas dispersion of flammable vapor resulting from release of cryogenic liquids like Liquefied Natural Gas (LNG) by the application of mesoscale meteorological models and local measurements. Additionally it highlights the importance aspect of local meteorology. Wind fields and temperature data derived from this system are compared with the meteorological field observations from the Doha International Airport (DIA). Different averaging periods are selected to demonstrate the variations in meteorological parameters. A case study is performed by taking meteorological inputs from DIA field observations and CALMET predictions of different averaging periods. The results of atmospheric details along with parameters like release rate, release duration, local geographic details and specific aspects of process equipment are used to estimate the extent of a vapor cloud. This extent is done by determining of Lower Flammability Level (LFL) of a dense gas vapor of LNG. The LFL estimates serve as key inputs to perform risk assessment of facilities handling Liquefied Natural Gas (LNG).

2. Materials and methods

2.1. Area of study

A specific area of study is the Ras Laffan Emergency and Safety College (RLESC) located in Ras Laffan, Qatar. The Ras Laffan City is situated along the northeast coast of Qatar overlooking the Persian Gulf. Its strategic location is at the center of the Persian Gulf, between the Far East and Europe on the international maritime shipping route, enabling hydrocarbon products to be efficiently transported to markets all over the world. Qatar, being a region of coastal peninsula, the prevailing winds is from the northwest direction from the Persian Gulf (Qatar Meteorology Department, 2014). The climate of Qatar can be described as subtropical dry, hot desert climate with low annual rainfall, very high temperatures in summer and a big difference between maximum and minimum temperatures, especially in the inland areas (Qatar Meteorology Department, 2014). The coastal areas are slightly influenced by the Persian Gulf, and have lower maximum, higher minimum temperatures and a higher moisture percentage in the air (Qatar Meteorology Department, 2014). As winds enter Qatar, they are directed outward to the ocean. Ras Laffan City provides infrastructure, facilities and services for the functioning of hydrocarbon industries utilizing gas reserves in the North Field which lies 80 km offshore Ras Laffan. The RLESC facility has been designed for fire training purposes, and includes 29 different props to train fire-fighters in different situations (Fig. 1). A meteorological analysis is performed here for cryogenic liquid spill experiments to be conducted in TP-5 of RLESC.

2.2. Meteorological model

To facilitate the comparison of model data and observed data hourly meteorological data is obtained from a meteorological station positioned at Doha International Airport (DIA, Station ID-411700, Latitude $-25^{\circ}25'$, Longitude $51^{\circ}56'7''$, Elevation -10 m). The location of this station is denoted as yellow (in the web version) circle in Fig. 2(a). The original station format is converted to a



Fig. 1. Ras Laffan emergency and safety college.

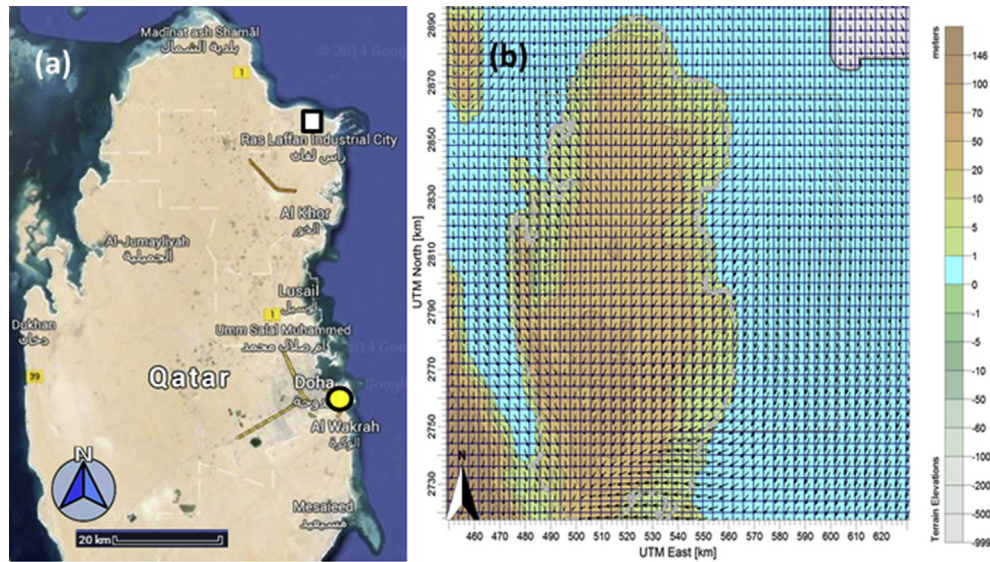


Fig. 2. (a) Rectangle represents area covered by Model data and circle represents the location of field observations. 2(b) Vector wind field with grid for Qatar Region.

readable format to facilitate comparisons between CALMET model and actual meteorological data.

The MM5-CALMET modeling system is employed to simulate the meteorological conditions of Qatar. It includes a meteorological processor (CALMET), a dispersion model (CALPUFF), and a post-processor (CALPOST). CALMET is a diagnostic wind field model with micro-meteorological modules for overwater and overland boundary layers (Scire et al., 2000). CALMET includes both diagnostic and prognostic wind field generators. CALPUFF is a multi-layer, multi-species non-steady-state Gaussian puff dispersion model that simulates the effects of time and space-varying meteorological parameters. CALPOST contains post processing programs for the output fields of meteorological data (Scire et al., 2000).

CALMET is first initialized with Mesoscale Model data (MM5) which is used for creating weather forecasts and climate projections. The Fifth Generation Penn State/NCAR Mesoscale Model (MM5) (Grell et al., 1994) is used for this purpose. The meteorological domain extended 441 km horizontally and 2708 km vertically with 11 vertical layers along the elevation. MM5 data with a 4 km spatial resolution are used as input to CALMET. The domain encompasses the Ras Laffan Industrial Area which is denoted as white rectangle in Fig. 2(a). Fig. 2(b) shows an arbitrary wind vector plot of Qatar along with the grid used in CALMET.

The MM5 is a prognostic wind field model with four dimensional data assimilation. The diagnostic wind field model in CALMET, called the CAL-MM5, extracts and converts MM5 meteorological data into a form compatible with CALMET. This format is usually a CALMET.dat file that contains hourly fields on winds and meteorological variables. The CALMET.dat file, an output file of CALMET, contains the three-dimensional gridded fields of u , v , and w wind components and air temperature, two-dimensional fields of Pasquill–Gifford stability class, surface friction velocity, relative humidity and precipitation rate. The wind fields produced by MM5 are either used as an initial guess field or combined with other data sources for the CALMET objective analysis procedure. The time averages of meteorological variables are reported in a convenient format to aid in the comparison of model and actual meteorological data.

2.3. Methodology

Default meteorological parameters like wind speed, wind

direction and temperature are considered for comparison between CALMET model data and DIA field data. Weather data required for risk analysis is usually defined by these three site-specific variables. Variations in wind speed and atmospheric temperature affect the size and extent of impact zones. While wind directions are important in mapping the risk contours surrounding the leak areas. The weather conditions along with the release conditions are generally used to generate a site specific hazard scenario. Other parameters such as the mixing height, cloud cover and solar radiation are not completely available in field observations. Although these parameters can be estimated, it may constitute a source of uncertainty due to the limited ability of all the required measured data. The data used here has been recorded for a period of three years starting from December 2009 to January 2013. It was previously studied that field measurements of meteorological variables collected for at least one year is required for predicting the weather conditions (Marx and Cornwell 2009). The three year data selected here provides a substantial insight on the annual variations present in meteorological parameters. The data for DIA station is retrieved from the National Climatic Data Center (NCDC). The data collected from Doha International Airport is 91.2% complete with missing data amounting to 8.8%. The DIA contains meteorological parameters collected at a frequency of 1 h. Different averaging periods are chosen to study the effect of temporal averaging on meteorological parameters.

2.3.1. Monthly averaging

In the monthly averaging, the meteorological parameters which are recorded in an hourly basis in monitoring stations are averaged on a monthly basis to obtain the monthly average. One average value is obtained for each month for three years. The average values obtained in MM5-CALMET model are compared with average values obtained in DIA station data to study the annual trend of meteorological parameters.

2.3.2. Daily averaging

In the daily averaging, the meteorological parameters which are recorded in an hourly basis in monitoring stations are averaged on a 24 h basis to obtain the daily average. The average values obtained in MM5-CALMET model are compared with average values obtained in DIA station data. The observed and predicted primary meteorological parameters namely wind speed,

atmospheric temperature, and wind direction, is discussed here owing to their importance in atmospheric dispersion. The daily averaged plots are generated for wind speed and temperature, for duration of three years. Wind rose plots are generated for wind direction variable.

2.3.3. Hourly averaging for diurnal analysis

In coastal regions like Qatar, the dispersion is bound to display differences during day and night. This is due to the significant diurnal variations that are brought about by the difference in sensible heat flux between land and water. Hence, a study on the variation of atmospheric parameters during day and night is performed. Hourly averages are obtained by averaging the meteorological parameters obtained for a particular hour for three years. For example, Jan 1st 2010, Jan 1st 2011 and Jan 1st 2012 600 h data is averaged to obtain one hourly average for 600 h for January month. Diurnal analysis is performed by plotting the meteorological parameters with respect to the hour of the day.

2.3.4. Validation

The performance of the model lies in validation with field observations. To test this performance of the MM5-CALMET model for analysis, performance statistics are commonly used (Chang and Hanna 2004). The model is bound to display a small error if the variability in the model solutions is equally compensated by the natural variability occurring in the observations. However this variability in observations and model predictions arise from different sources. To account for variations and establish the performance of model prediction, a factor of 2 is widely used. According to this statistical performance measure, a model is deemed to be good if the majority of the solutions fall within a factor of 2 ($\pm 33\%$) of the observations. The factor of 2 is a fraction of data that satisfies $0.5 \leq C_p/C_o \leq 2$ where C_p is the model prediction and C_o is the observation.

3. Results and discussion

An evaluation of the performance of the MM5-CALMET meteorological output is conducted by comparing the model predicted meteorological parameters with the observed measurements from monitoring stations within the domain of the study. The averaging periods are on a monthly basis, daily basis and hourly basis for diurnal analysis where average values of parameters are calculated per month, per day and per hour. The result for wind speed, wind direction and temperature parameter is discussed in Section 3.1, 3.2 and 3.3 respectively. Each section describes the role of each meteorological parameter in dense gas dispersion. A case study illustrating the effect of variation of meteorological parameters on the lower flammability limits resulting from flammable dense gas release of LNG is provided in Section 4.

3.1. Wind speed

The wind speed plays an important role in influencing atmospheric dispersion. The intermixing of air and dense gas can lead to turbulence generation and subsequent turbulent mixing which finally leads to cloud dilution. The cloud dilution is important in determining the distances up to which the cloud will travel and its potential effect on people if it is flammable. Dense gas clouds, like that of LNG are negatively buoyant and they tend to slump to the ground when the wind speed is low. As a result of this process, the dilution with air is reduced (Meroney and Neff, 1982).

Fig. 3 provides a comparison of annual patterns of wind speed between DIA dataset and MM5-CALMET. The solid lines represent the field data and the dashed lines represent the model data. The

MM5-CALMET model prediction is close to the wind speed data of DIA dataset. Wind speeds values are found to be moderate with magnitude ranging from 3 to 5 m/s. The simulated wind speed is in generally 0.5 m/s higher than those observed in the year 2010 and lower by 0.5 m/s in the year 2011. Both under-prediction and over-prediction is obtained for year 2012. It is important to note that the overall pattern of the wind variable is well captured by the CALMET model. It can be also observed from Fig. 3 that the variations of wind speed in the years 2011 and 2012 is found to be minimal in the second half of the year, when compared to simulated 2010 values, which had variations throughout the year. These wind variations are not simulated completely by the model due to coarse resolution of the grid. One of the possible reasons is that elevation of 4 km grid resolution might have been less than the actual terrain height. Hanna and Yang (2001) raised similar concerns about modeling the strength of elevated inversions of wind speed. The performance of capturing such small changes in wind speeds are largely related to the vertical resolution of the simulations.

Fig. 4 depicts the comparison of the CALMET average wind speed per day for duration of three years with an observed average wind speed for Doha International Airport (DIA). The scatter plot employs a total of 1095 data points. The reported wind speeds are within an average value of $3.7 \pm 0.27 \text{ ms}^{-1}$ for both CALMET model and DIA field values. The FAC2 value for DIA field data is found to be 0.98. For the DIA monitoring station, the observed values of wind speed are higher than the predicted values for most of the months. Over-predictions are observed for the year 2010 and under-predictions are observed for the year 2011 and 2012. The variations are attributed to the missing data present in field observations, which can potentially lead to inhomogeneity. It is to be noted that missing data is not present in CALMET prediction as wind speed predictions are available for all time intervals. However, missing data is present in the DIA data which can contribute to inhomogeneity. Such inhomogeneity in field data can arise due to changes in instrumentation and changes in station exposure surrounding the station. These are generally caused by factors like variations in sea surface temperature, obstruction by new buildings and nature of the terrain (Bichet et al., 2012).

Hourly averages for diurnal analysis of a particular month are obtained by plotting the DIA and MM-CALMET meteorological parameters with respect to the hour of the day. Fig. 5 (a)–(d) shows the variation of wind speed during the day calculated for a three year period for selected months March, June, September and December. The chosen months are representative of the seasonal variations of Qatar. The rest of the months are found to follow a trend similar to the chosen months. The solid lines represent the observation and the dotted lines represent the model values.

The model generated wind speed values reflect an average value and very fewer variations are observed. However, the model values are comparable to station data for night time wind speeds for all months. The wind speeds are generally reasonably close in magnitude to the observations. The bias observed in night hours (1800–600) are found to be lesser in months March and June and greater in September and December months. The wind speed values obtained in MM5-CALMET model show less variations as it averages the wind speeds in day time and night time conditions. However, distinct variations in wind speed are observed in DIA observation. The low wind speed, which is observed during the night time, is averaged to a value closer to MM5-CALMET prediction, whereas the relatively high wind speed values obtained during day time caused significant differences in averaged values between model and data.

One can observe from diurnal plot of observed wind speeds that as the day length increases, wind speed also increases and reaches a maximum value at noon. Once a maximum value of wind speed is

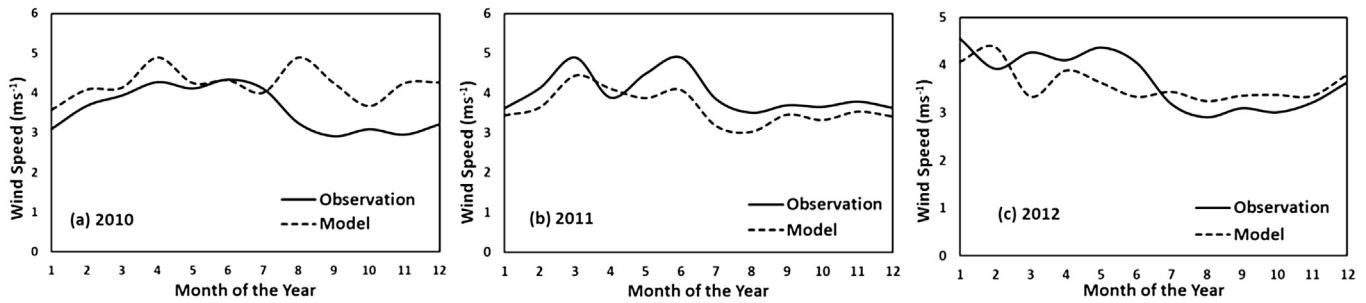


Fig. 3. Comparison of monthly averaged values from observation and predicted wind speeds for years (a) 2010 (b) 2011 and (c) 2012.

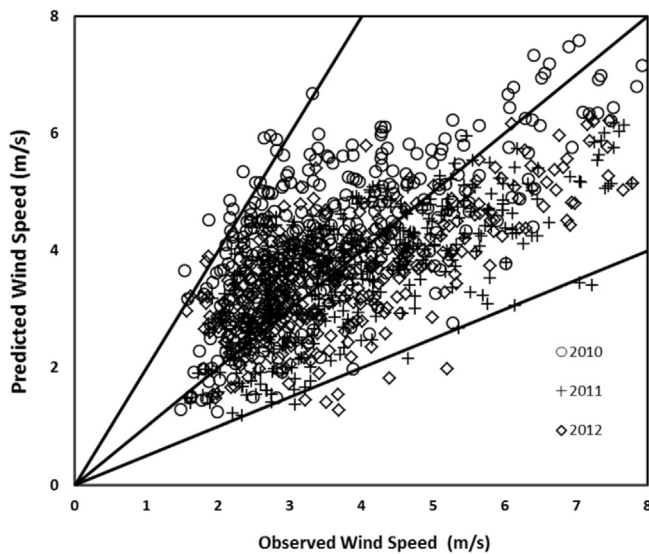


Fig. 4. Comparison of observed and predicted wind speed on a daily basis.

reached, it decreases to a minimum value at night. This is due to differential surface heating (Robe et al., 2000). Solar heating during the day causes air near the land surface to heat more rapidly than the air over water and causes it to rise and be replaced with air that was once present above the cooler body of water. The increased speeds usually occur around 1200–1600 h. A dense gas release during this time can possibly lead to dispersion of cloud to large distances away from the leak source. After afternoon, the land cools off faster than the ocean due to differences in their heat capacity, which forces the dying of the day time sea breeze. If the land cools below that of the adjacent sea surface temperature, the pressure over the water will be lower than that of the land, setting up a land breeze. The land breeze pushes cooler air from the land to sea during the night and these results in lower wind speeds in the land. The land breeze will die once the land warms up again the next morning. This is observed during the day time from 400 to 700 h. A dense gas release during these hours is likely to result in short dispersion distances due to the decrease in wind speeds. During night hours, one can expect the dense gas cloud to be present near the leak area for a long duration of time. Since Qatar is a coastal area, meteorological conditions such as sea/land breeze recirculation can significantly affect the dispersion of dense gas. Steyn and McKendry (1988) also suggested that the models have difficulty with sea breeze-land breeze transition periods. Any flows driven by local thermal gradients, such as sea breezes, become extremely important during periods of weak synoptic pressure gradients. Under these conditions, high spatial resolution and good

characterization of the surface radiation budget is very important. However, these meteorological conditions are highly complex in a spatial and temporal sense and are difficult to capture this completely by the model. As indicated earlier, for the similar diurnal case, the observations differ significantly whereas the model predictions show similar diurnal variations. The reason for low variations in model can be attributed to the different exposure of DIA station and MM5-CALMET model location. Their separation distance is larger than (about 20 times) the selected grid length (4 km). In such situations, higher horizontal resolution is required in the model in order to capture these scale based variations.

3.2. Wind direction

The wind direction specifies the direction of dense gas cloud travel. An important part of risk analysis is to determine whether an accidental release of dense flammable gas can cross the boundaries of the facility (NFPA, 2013) and (49-CFR-193, 2013). The cloud can be carried by wind for long distances which can pose hazards for both environment and human health. The prevailing wind direction in Qatar is north and northeast during the first half of the year. This trend changes to north and northwest during the second half of the year. In the western direction of RLESC location, there is desert and dispersion in this direction is less likely to pose any major risk to the facility. When the predominant wind direction is north or northeast, the wind is directed in the southern and southwestern direction which can travel to the training props present in the facility. A dispersion of flammable vapors directed in the eastern direction may pose risk, as classrooms and control rooms are present in that direction.

Fig. 6 shows the comparison of wind direction obtained from DIA dataset and CALMET/MM5 model. The comparison with DIA dataset is reasonable. Wind direction is predominantly to the Southsouthwest during the first half of the year. The observed wind direction changed from the Southsouthwest to southeast and finally to east gradually during July and August. The simulated wind direction followed a similar trend to the observation. However, the simulated wind direction is initially in southsoutheast direction and then shifts to southeast and eastern direction. A maximum difference of 50° is observed between observed and modeled values. The annual trend of wind speed is well captured by the model, however, the values of wind direction showed some disagreement. The disagreement of wind direction values can be attributed to the location of DIA station. DIA station is located near shoreline indicating that wind speeds are subjected to changes from the oceans. The model did not reproduce the rapid shift in wind flows that are considered to be sub-grid scale phenomena with 4 km grid spacing used in this model.

To analyze the direction of vapor cloud travel, wind roses are chosen to describe the wind direction. The wind roses required for

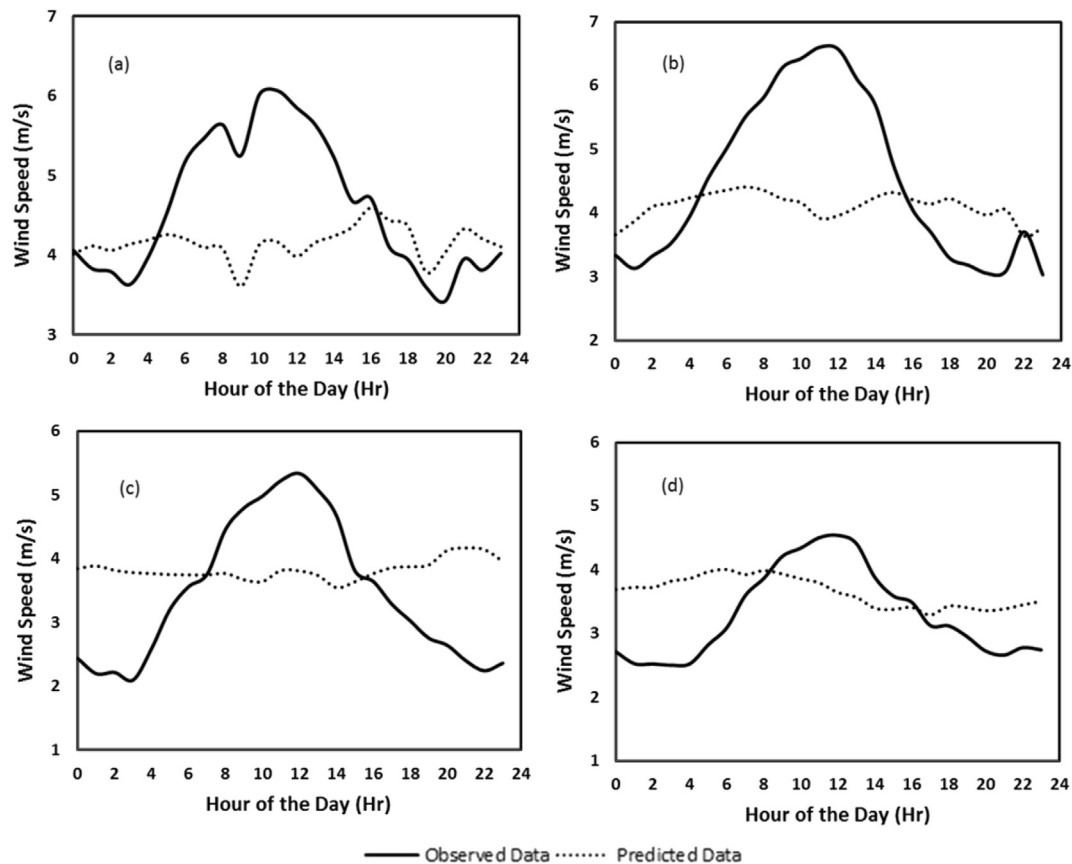


Fig. 5. Comparison of observed and predicted Wind Speed for Months (a) March (b) June (c) September and (d) December.

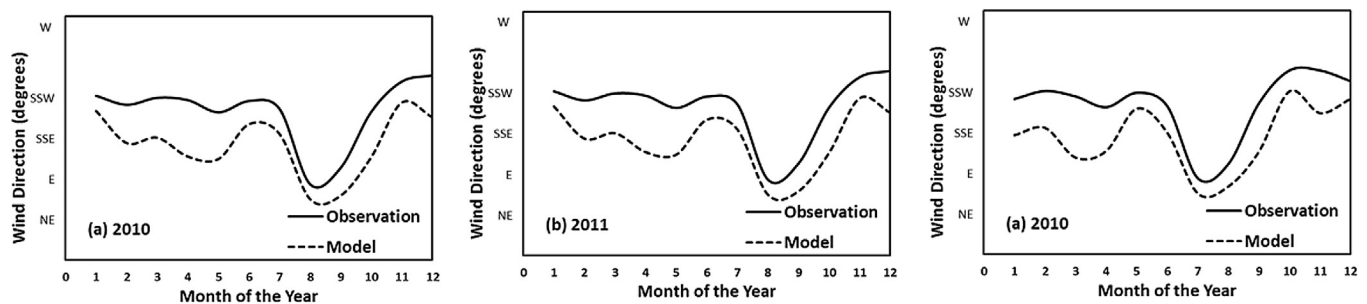


Fig. 6. Comparison of monthly averaged values from observation and predicted wind directions for years (a) 2010 (b) 2011 and (c) 2012.

analysis are generated using WRPlot (Thé et al., 2000). The WRPlot processes various formats of meteorological data including CALMET binary data. A wind rose is usually generated from time-series meteorological data where input is transformed into a joint frequency distribution consisting of a ratio between wind direction and speed according to atmospheric stability. The percentage of wind speed range for each wind direction is then displayed using a concentric scale of frequencies (*i.e.*, each circular ring corresponds to a particular frequency, in percent). The frequency data is further distributed graphically based on the number of wind directions specified. It provides one of the most powerful graphical evaluation procedures of evaluating wind speed and direction at specific site locations of both CALMET model input data and model output data. The number of time periods used for generating wind roses is 8760 for each year. The percentage of calm winds varies in each category and is shown in a tabulated format in Table 1. A calm wind is a

condition where the wind speed is below 0.5 m/s. The calm winds are recorded in meteorological data and are usually excluded. Separate treatment of calm wind is required and is not considered here as it is beyond the scope of this paper.

Fig. 7 shows the wind rose plots generated for DIA and CALMET model prediction data for the three years from 2010 to 2012. Dispersion towards the direction in which desert is present is considered a favorable condition owing to the reduction in

Table 1
Number of calm winds observed in each year.

Year	Calm winds (DIA)	Calm winds (CALMET)
2010	165	153
2011	125	166
2012	137	129

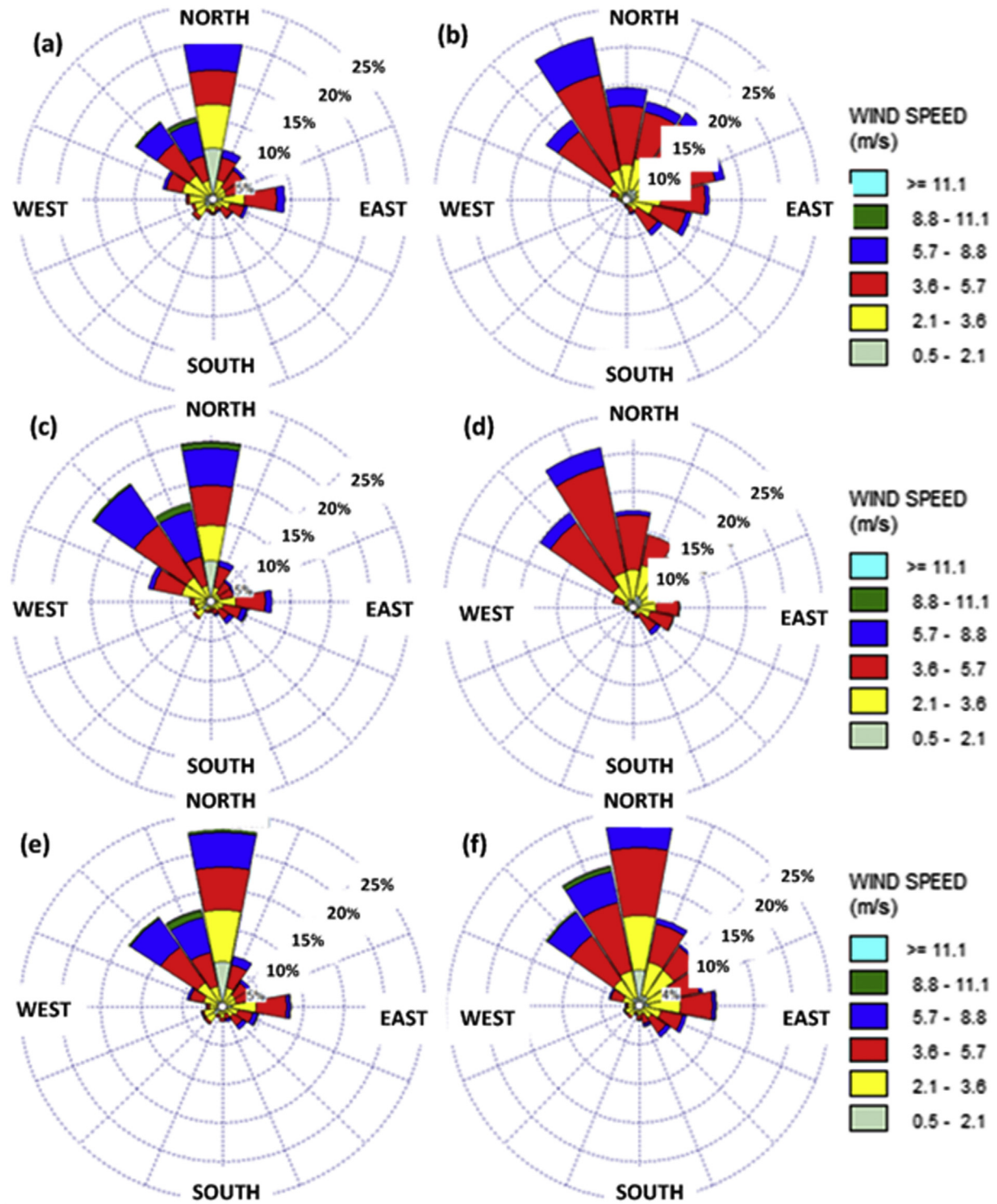


Fig. 7. Comparison of Wind Roses (a) DIA Station data for year 2010 (b) CALMET Model data for year 2010 (c) DIA Station data for year 2011 (d) CALMET Model data for year 2011 (e) DIA Station data for year 2012 and (f) CALMET Model data for year 2012.

occurrence of a hazard. This trend is observed in the western direction. Similarly, high wind speeds in the direction of desert are favored for dispersion, as the gases are dispersed from the area of discharge sooner.

The predominant direction at DIA is north-northwest and the predominant wind direction is northwest in the CALMET model. One can observe that the CALMET model has higher magnitude of

wind spread in the east and southeast Direction, but the frequency of wind speeds in these direction are found to be low in DIA station wind roses. The lower frequency in DIA station wind roses can be attributed to the wrong positioning of common industrial meteorological stations. This can lead to meteorological measurements that are extremely biased causing a misconception in wind roses. The wind directions are evenly distributed in Westnorthwest,

Northwest, Northnorthwest, Northnortheast, Northeast and East-northeast direction in the CALMET model. However winds are observed to blow in Westnorthwest, Northwest, Northnorthwest, North direction in the DIA model. The wind speeds obtained from DIA data demonstrate a high frequency of wind speed ranging from 5.7 to 8.8 m/s and variability in predominant wind direction to North and NorthWest. Similarly, the wind speed frequencies obtained from CALMET are more concentrated in the range 3.6–5.7 m/s with variability in the predominant wind direction to North and NorthNorthWest.

The wind roses generally showed poor agreement for wind direction and wind speed frequency. From the wind roses, one can also observe that the predominant wind direction captured by CALMET model is less representative of the DIA dataset. However, one can conclude that a close estimate of wind direction can be obtained from the CALMET model. This sub-optimal performance is attributed to different sampling locations of DIA field data and model predictions. A site specific monitor data obtained directly from Ras Laffan will provide better results as the distance and variations in topographical features will be less between station and model data.

Fig. 8 (a)–(d) shows the variation of wind speed during the day calculated for a three year period for selected months March, June, September and December. The time series of surface wind direction exhibits a steady directional shift from the southerly to westerly mean flow in the month of March. The observed wind directions changes from westsouthwest direction to west direction in the day and southsouthwest in the night, but the simulated wind direction remains between south and southsouthwest throughout the 24 h period. Such directional variations can be studied in a detailed perspective in the diurnal plots. During the June Month, the diurnal variations are found to be very low. The winds blowing toward

northern and western direction are conducive for dense gas release experiments as the winds are directed towards the ocean and desert respectively. The occurrence of hazards in these directions is low. There also appears to be a fairly unusual diurnal variation in wind direction from east to south in September month during the hours 1600–2200 h. This shift is detected earlier in the model, when compared to the DIA Station data. Of interest is that, in every month, peaks are observed at regular intervals which suggest the likely shift in wind directions of land breeze and sea breeze. During the December month, the wind direction steadily drifts from Westsouthwest to west and from west to South-southwest twice during the day. Hence winds will be directed toward the eastern and north eastern direction. A comparison of wind direction observed in station and model data reveals that the trend in variation of wind direction is well-captured by the model to certain degree of accuracy but a bias is observed between model and predicted wind direction.

3.3. Temperature

A direct dependence of atmospheric temperature on a vapor cloud is not seen; rather its implicit effect on other parameters is observed for dense gas dispersion. The temperature has dependence on mixing height and surface layer parameters like roughness. Vertical temperature gradient measurements are used for classifying stability in the atmospheric boundary layer. The temperature gradients are further used to get estimates of the plume rise for stable conditions. For application of temperature to dense gas dispersion simulations, the accuracy and resolution of the temperature measurements are important, as changes in temperature are small in regions like Qatar.

Fig. 9 shows the annual variation of air temperature for DIA

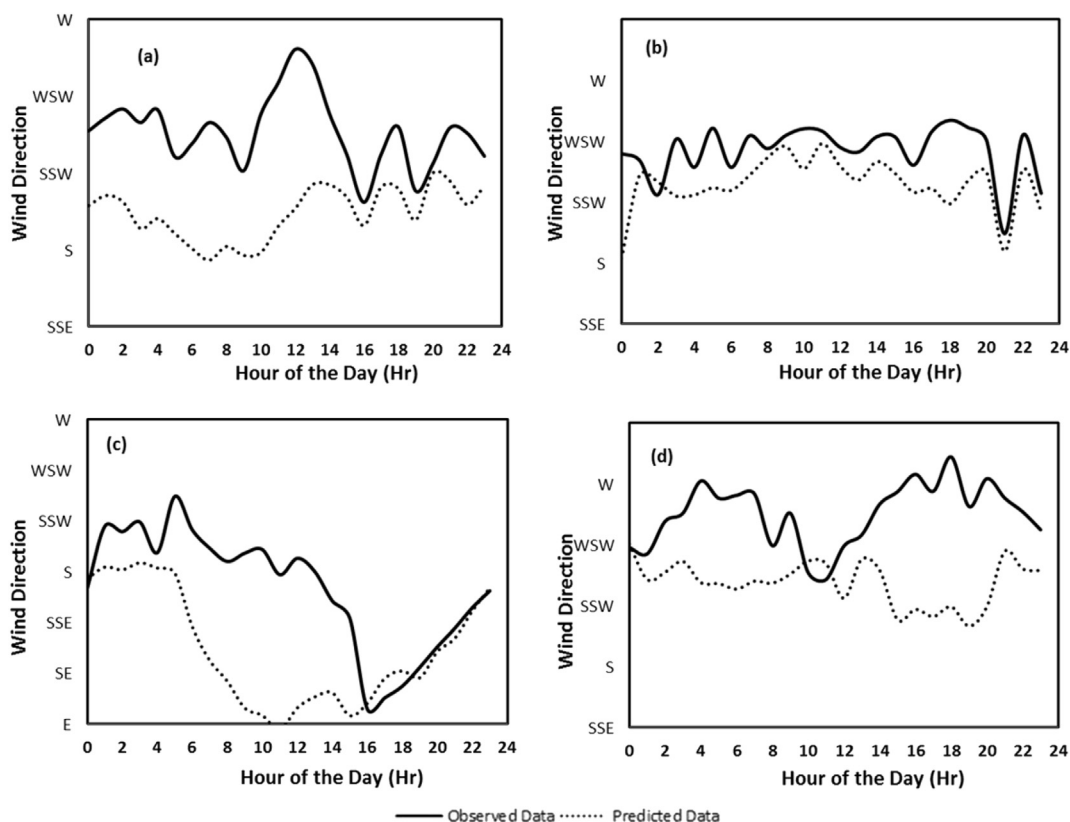


Fig. 8. Comparison of observed and predicted Wind Direction for Months (a) March (b) June (c) September and (d) December.

dataset and model. The temperature increases gradually every month from 292 ± 3 K to a maximum value of 315 ± 5 K during July month. In general, the numerical results for the winter situation (Nov–April) are better than the summer case (May–Oct). During summer time strong thermally induced wind systems dominate the flow field. In contrast the winter situation is characterized by a weak diurnal variation and the wind field is determined by terrain features. This weak variation of temperature during winter is well captured by the model. The DIA observations are in good agreement with the model values. Additionally, the annual trend is also well simulated by the MM5-CALMET model.

A good prediction is also observed in the temperature variable in daily averaging. The number of data points chosen for constructing the scatterplots is similar to that of wind speed variable. The predicted values of temperature are much closer to the observed values for DIA site. However, the DIA monitor data showed a certain amount of deviations in all three years. This trend is shown in Fig. 10. The temperature varied from 290 to 310 K for all three years, however outliers are also observed in the data. This might be attributed to the changes in exposure surrounding the station causing deviations from the expected behavior. The temperature variable shows very less variations when compared to wind speed and uniform trend is obtained for all three years. A FAC2 value of 0.99 is observed for the temperature variable.

Fig. 11 (a)–(d) shows the variation of temperature during the day calculated for a three year period for selected months March, June, September and December. The model prediction for air temperature is in good agreement as very less deviations are observed between model and field data. It can be observed that the temperature variable is less affected by spatial variations present in Qatar Region. A robust prediction of temperature variable is favorable for dispersion. A dense gas cloud formed by release of LNG is likely to undergo two stages of dispersion - increase in vertical momentum of the cloud due to vaporization and entrainment of air into the cloud leading to mixing (Koopman et al., 1989). The shift from one stage to another is influenced by the atmospheric boundary layer which is described by wind and temperature profile. Unstable conditions occur during the day as the ground is heated by the sun. The warm ground leads to unstable density stratification. A release during the day time will lead to dilution which is brought about by thermal effects of the atmosphere. This is possible during time periods from 400 to 1500 h. The temperature is bound to reach a maximum value during this time period. This trend can be observed in Fig. 11 (a), (b) and (d). Neutral atmospheric condition occurs in the evening when the heat transfer from the ground is very small, nearly zero. A release during the evening can lead to prolonged dispersion as neither thermal effects and buoyancy effects are dominating the cloud. This is observed in time periods from 1500 to 2200 h. During a stable stratification, the convection and conduction cools the atmospheric boundary layer

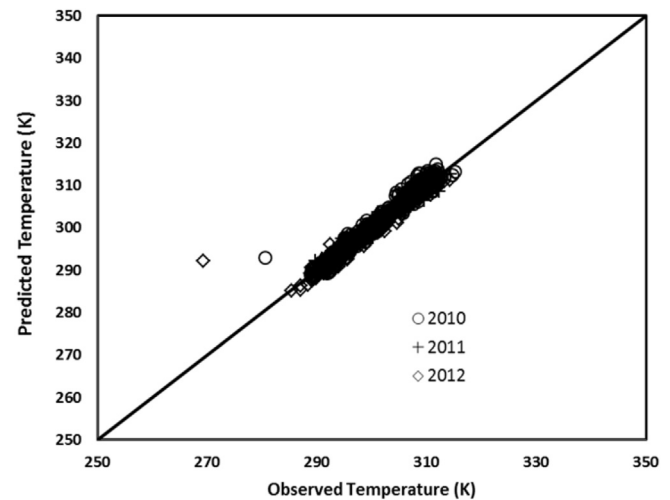


Fig. 10. Comparison of observed and predicted temperature on a daily basis.

leading to limited vertical mixing rates in the atmosphere, i.e. the dilution is minimal and usually happens during the night. A release during the night leads to very low dilution of the cloud as the atmosphere is cool and buoyancy differences between dense gas cloud and air dominate rather than temperature differences. This can be observed in time periods 2200 to 400 h. The MM5-CALMET model is found capable of simulating the temperature changes that occur due to shifts in atmospheric boundary layer.

A summary of validation of observations with simulated data is expressed in Table 2. Table 2 shows the FAC2 values of different meteorological parameters for different averaging periods. The prediction is found to be best for the temperature variable considering all the averaging periods. The prediction is good for wind speed variable with values of FAC2 close to 1. However, the prediction of wind direction variable using MM5-CALMET is moderate. Overall, the hybrid approach of using a coarser resolution prognostic model in combination with a diagnostic model is found promising in simulating the meteorological fields. This approach makes it feasible to capture at least some of the effects of small-scale topographical features. However, there are issues identified with grid resolution and distance between sampling location and model location. From this study one can conclude that grid resolution lesser than 4 km and fine vertical resolution are required for providing more realistic results.

4. Case study

A case study is performed to study the effect of weather variations on lower flammability limit of LNG release for weather

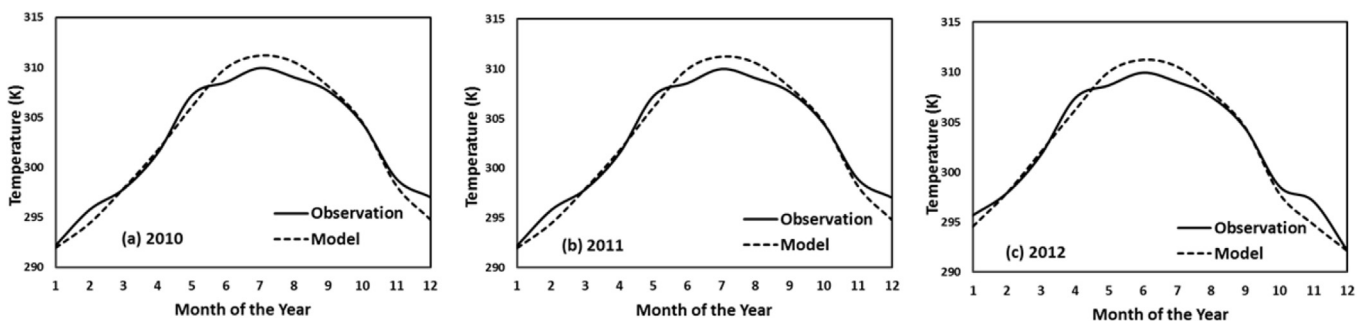


Fig. 9. Comparison of monthly averaged values from observation and predicted wind directions for years (a) 2010 (b) 2011 and (c) 2012.

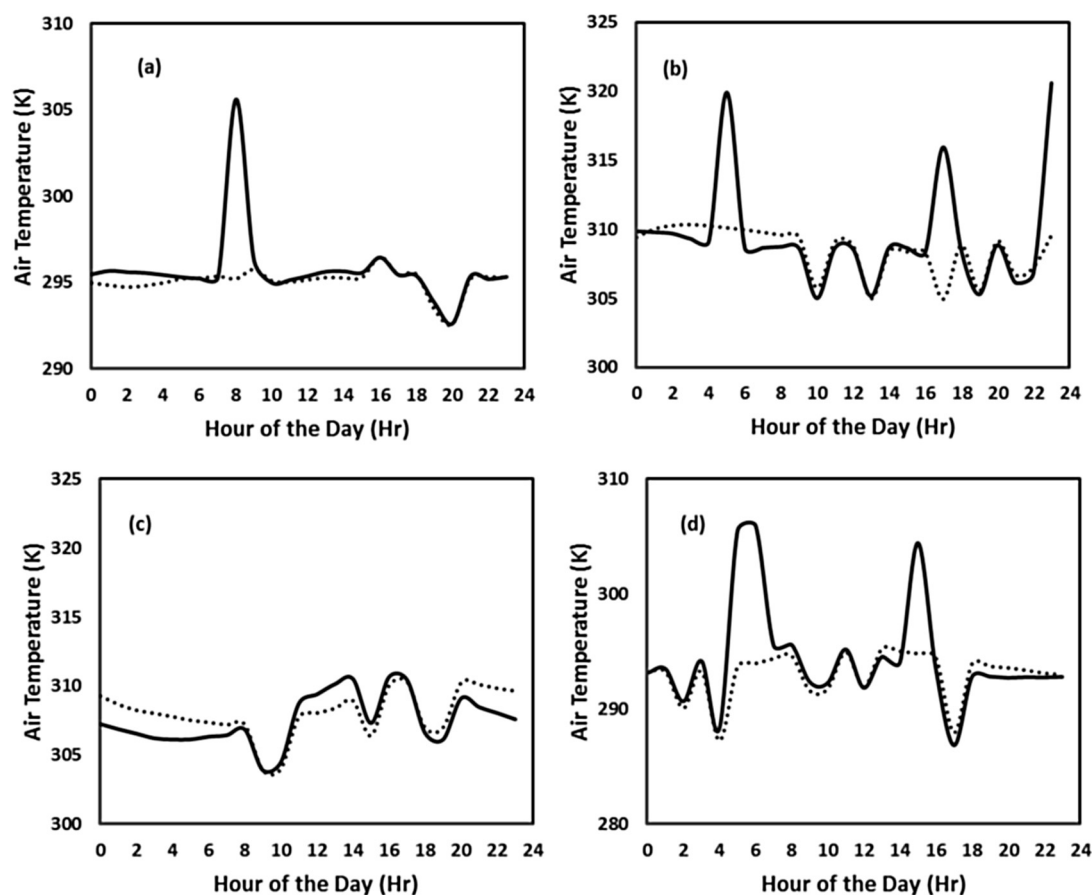


Fig. 11. Comparison of observed and predicted Air Temperature for Months (a) March (b) June (c) September and (d) December.

Table 2
Summary of FAC2 values for different averaging periods.

Averaging period	Meteorological Parameters		
	Wind speed	Wind direction	Temperature
Monthly	1.03	0.78	1.00
Daily	0.98	0.94	0.99
Hourly	1.09	0.88	1.00

conditions with different averaging periods. The area chosen for study is the LNG training prop (TP-5) present at the Ras Laffan Emergency and Safety College, Qatar. The release scenario is a continuous release of LNG at a rate of $0.013 \text{ m}^3 \text{ s}^{-1}$ (200 gpm) for duration of 600s. As it is not feasible to look at all potential spills within the plant, a spill representative of a medium scale spill is investigated. Therefore, they represent a reasonable estimate on spills that could occur in a particular facility. This is similar in concept to the “accidental leakage design spill” that is used to evaluate the consequences of an accidental release of cryogenic liquids like LNG (NFPA, 2013) and (49-CFR-193, 2013). Two different sets of meteorological input from DIA field data and MM5-CALMET prediction are selected. The selection of meteorological parameters involves parameters from different averaging periods like monthly, daily and hourly from DIA field data and MM5-CALMET model data for the month of September. The selection of day and month are on an arbitrary basis to investigate the rigorosity of the model. The corresponding stability classes with highest occurrence during September are chosen from DIA data and MM5-CALMET model respectively.

The release scenarios involved the four weather variations corresponding to different averaging periods. All the other release variables are held constant in order to provide a comparative analysis of meteorological parameters on lower flammability limits. A summary of weather parameters for DIA meteorological observation (Data) and MM5-CALMET meteorological prediction (model) along with the lower flammability results is shown in Table 3. An LFL contour from flammable dispersion analysis is determined using PHAST 6.7 (DNV (Det Norske Veritas), 2012).

From Table 3, one can observe a maximum difference of about 50% is observed between LFL values obtained from DIA and CALMET input data for night averaged meteorological input. This difference highly suggests the geographical variations leading to different weather conditions in each location. This is likely due to the difference in wind speeds and wind direction between DIA meteorological observation data and MM5-CALMET prediction. The wind speed values obtained in CALMET model show less variations as it averages the wind speeds in day time and night time conditions. However, distinct variations in wind speed as observed in DIA field observation for day and night time. This has a pronounced effect on the LFL values for night time conditions. However, it showed a small difference for the day time conditions. A difference of 13% is observed between data and model LFL distances. In the day time weather conditions, one can observe the low value of LFL distance with high wind speed. This is due to the fact that, with increased wind speed, vapors get dispersed to safe flammability levels closer to the leak area.

In Table 3, one can also observe that the LFL values obtained

Table 3

Summary of case study showing the meteorological parameters and LFL distances.

Month	Wind speed ms^{-1}		Wind direction		Atmospheric temperature (K)		Stability class	LFL distance (m)	
	Data	Model	Data	Model	Data	Model		Using DIA meteorological data	Using CALMET meteorological Model
Monthly average	6.5	4	S	ESE	307	310	C	32	40
Daily Average	3.7	5.6	S	NE	307	309	C	56	44
Day Time	5.5	4	S	E	312	310	C	45	40
Night Time	2.2	4	SSW	S	307	310	F	61	91

during the night time is higher than day time LFL values. This is likely due to the stable atmospheric conditions which inhibit dispersion. In night time, one can also observe the pronounced effect of temperature on the LFL values. The high night time temperature and significant wind speed values leads to increase in dilution of cloud. The DIA input data for night time weather conditions, obtained the highest LFL of 91 m which is twice as high monthly, daily and day averaged LFL values. Similarly, high wind speed values are observed during the day time. High wind speeds is a contributing factor to the dispersal of vapors. This is observed in LFL values of DIA data using monthly averaged meteorological inputs.

Fig. 12 shows the variation of $\frac{1}{2}$ LFL contours and the downwind direction of LNG vapor dispersion obtained for September month night time condition. There exist a variation in wind speed, wind direction and temperature between DIA field data and MM5-CALMET model data. However, the stability class is found to be similar. An LFL prediction results in different distances with a variation of about 50%. From the Fig. 12, one can also observe the different direction of dispersal of vapors. In the DIA based LFL, wind blows from southwestern whereas in MM5-CALMET based LFL prediction, wind blows from southern direction. Similarly a simulation of LFL, based on MM5-CALMET values for day time, daily averaged or monthly averaged values can lead dispersion of dense gas in eastern and northeastern direction where classrooms and office building are present. This scenario can potentially pose significant risk to the people. Such cases of model predicted values clearly demonstrate the issues that arise in a risk analysis study when “wrong” meteorological data could be used. These factors highlight the importance of incorporating proper meteorological variations into risk analysis. A single set of weather conditions, not taking into account the variations for a particular region, can result in under-prediction or in the worst case, wrong prediction of LFL distances and direction of cloud travel, which can potentially increase the risk associated with flammable dense gas vapors.

5. Conclusion

In this paper, a screening of meteorological data is performed for a three year period using primary meteorological parameters like wind speed, wind direction and temperature. The results of this study indicate that utilizing the coarsest prognostic meteorological model output in a diagnostic model provides a feasible option for generating meteorological inputs for risk analysis. This methodology proves beneficial in cases where there is no QA/QC meteorological data for the study area or when the wind flow pattern is complex e.g. away from coastal areas. The monthly averages are found to replicate the annual patterns of meteorological parameters. However difference in observation and model are observed for wind speed and wind direction variable. The daily averages obtained from the model are observed to be in good agreement with the station data for wind speed and temperature. For diurnal analysis, the model is found capable of mimicking the temperature of a given location, but not wind speed and direction. The prediction of wind direction parameter using MM5-CALMET is found to be moderate. The issues are attributed to grid resolution of vertical and horizontal layers of CALMET and MM5 model respectively. From this study one can conclude that grid resolution lesser than 4 km and fine vertical resolution are required for providing realistic results.

A case study illustrating the effect of variation of meteorological parameters on the lower flammability limits (LFL) resulting from flammable dense gas release of LNG is undertaken. The analysis of the case study illustrated the importance of different averaging periods on LFL limits. A better performance is expected for a site specific validation where sampling location is closer to the location of the model. On the other hand, this case study clearly demonstrates the issues that arise in a risk analysis study when “wrong” meteorological data could be used. In any case, further studies should include long term site specific observation data with sufficient spatial resolution to adequately characterize ambient atmospheric conditions. This is useful particularly in areas like Qatar

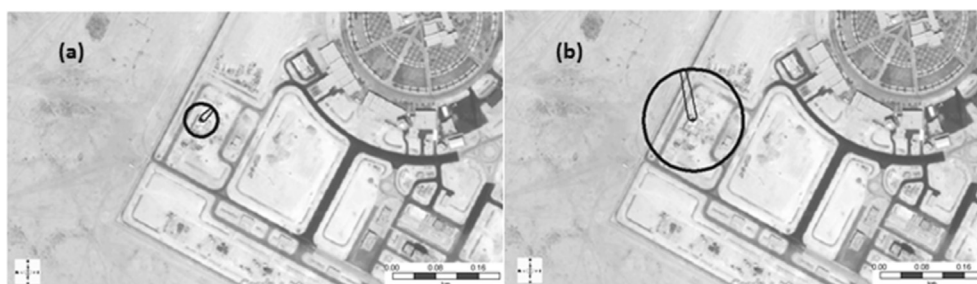


Fig. 12. Comparison of LFL distances and downwind direction for (a) DIA field data (b) CALMET prediction for night time condition.

where the meteorological conditions exhibits large spatial and temporal variations due to complex meteorology and variable land use (e.g. coastal areas). The results of this study indicate that utilizing the coarsest prognostic meteorological model output in a diagnostic model provides an effective option for generating meteorological inputs for dispersion studies.

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